

Evaluating VGG16 and ResNet50 for Simultaneous Skin Lesion Diagnosis and Age Estimation

Mostofa Seum*, Md. Sakib Hasan*, Emon Biswas*

*Department of Computer Science, American International University - Bangladesh, Dhaka, Bangladesh
{23-50299-1, 23-50316-1, 23-50174-1}@student.aiub.edu

Abstract—This project explores how well deep learning models can handle two medical imaging tasks at the same time: identifying skin lesion types and estimating patient age. We used the ISIC 2019 dataset, which contains 25,331 dermoscopic images across eight diagnostic categories. Two widely used neural network architectures, VGG16 and ResNet50, were trained for both classification (lesion type) and regression (patient age). We applied data augmentation, class weighting, and a careful data split to keep results fair and reliable. To understand what the models focus on, we generated visual heatmaps using Grad-CAM and Grad-CAM++. Results show that ResNet50 outperformed VGG16 on both tasks, achieving 68.82% accuracy for diagnosis and an average age prediction error of 10.34 years. The heatmaps confirm that classification models focus on lesion areas, while age prediction is more challenging because age is not directly visible in the images. This work demonstrates that modern CNNs can support dual medical tasks, with ResNet50 offering a good balance of accuracy and efficiency.

Index Terms—skin lesion, ISIC 2019, VGG16, ResNet50, age prediction, Grad-CAM, deep learning, medical imaging

I. INTRODUCTION

Skin cancer is a major health concern worldwide. Early detection can greatly improve treatment outcomes. Doctors use dermatoscopes to capture detailed images of skin spots, and in recent years, artificial intelligence has shown strong ability to analyze these images automatically.

Most existing research focuses only on diagnosing the type of lesion. However, patient information like age can also be clinically useful. This project tests whether a single model can learn two tasks at once: classifying the lesion type and predicting the patient's age directly from the image.

We used the ISIC 2019 dataset, which includes over 25,000 labeled skin images and associated patient metadata. Two well-known convolutional neural networks were tested: VGG16, a classic deep architecture, and ResNet50, a modern design that uses skip connections to train more effectively. Both models were trained under the same conditions to ensure a fair comparison.

The main goals of this project are:

- Build a system that performs both lesion classification and age estimation
- Compare the performance of VGG16 and ResNet50 on both tasks
- Generate visual explanations to show which image regions guide the models
- Use a careful, leak-proof data split to guarantee trustworthy results

II. RELATED WORK

The ISIC 2019 dataset has become a standard benchmark for skin lesion analysis. It builds on earlier collections like HAM10000, which proved that large, multi-source dermoscopic datasets could train reliable machine learning models.

VGG16 showed that stacking many small convolutional filters improves image recognition. ResNet50 solved a common problem in deep networks: as networks get deeper, they become harder to train. ResNet fixed this by adding shortcut connections, allowing much deeper networks to learn effectively.

Recent studies have tested newer models like EfficientNet and vision transformers on dermatological data. Some works also combine image features with patient metadata. However, most still focus only on diagnosis, leaving age prediction from images largely unexplored.

Interpretability is critical in healthcare AI. Grad-CAM creates heatmaps that highlight important image regions by using gradient information. Grad-CAM++ improves this by applying higher-order gradient weights, producing sharper and more precise maps. We used both methods to verify that our models focus on clinically relevant areas.

III. DATASET AND TASK FORMULATION

A. Dataset Description

We used the public ISIC 2019 dermoscopic image collection. It contains 25,331 images labeled into eight categories:

- Melanocytic nevi (NV)
- Melanoma (MEL)
- Basal cell carcinoma (BCC)
- Actinic keratoses (AK)
- Benign keratosis (BKL)
- Dermatofibroma (DF)
- Vascular lesions (VASC)
- Squamous cell carcinoma (SCC)

Patient age was also available for most images. We removed 437 records with missing age values, leaving 24,894 samples for the regression task.

Class Code	Full Name	Approx. Count
MEL	Melanoma	~4,522
NV	Melanocytic Nevus	~12,875
BCC	Basal Cell Carcinoma	~3,323
AK	Actinic Keratosis	~867
BKL	Benign Keratosis	~2,624
DF	Dermatofibroma	~239
VASC	Vascular Lesion	~253
SCC	Squamous Cell Carcinoma	~628

Fig. 1: Number of images per lesion class. The NV class is much larger than others, creating a strong class imbalance that affects model training.

Figure 1 shows the class distribution. The dataset is highly unbalanced, with NV making up the majority. This makes it harder for models to learn rare classes without special handling.

B. Data Splitting

To prevent data leakage, we split the data using lesion-level grouping. All images from the same lesion were kept in one set only. We used stratified group splitting to maintain class balance across splits:

- Training: 18,288 images (73%)
- Validation: 4,586 images (18%)
- Test: 2,457 images (10%)

No lesion IDs overlapped between splits, ensuring a clean and fair evaluation.

IV. METHODOLOGY

A. Image Preparation

All images were resized to 224×224 pixels to match the input size expected by VGG16 and ResNet50. Colors were normalized using standard ImageNet values.

Training images were randomly flipped, rotated ($\pm 15^\circ$), and slightly adjusted in brightness and contrast to help the model generalize. Validation and test images were only resized and normalized.

B. Age Normalization

For age prediction, we normalized values using the training set mean (54.66 years) and standard deviation (17.86 years):

$$\text{age}_{\text{norm}} = \frac{\text{age} - 54.66}{17.86} \quad (1)$$

Predictions were converted back to real age values before evaluation.

C. Classification Task

The models output eight probabilities, one for each lesion type. We used weighted cross-entropy loss to give rare classes more importance during training. This prevents the model from simply guessing the majority class.

D. Regression Task

For age prediction, the models output a single number. We used mean squared error (MSE) loss on the normalized age targets.

E. Model Training

Both networks started with weights pretrained on ImageNet. We froze early layers and fine-tuned the later layers to adapt them to skin images.

Training settings:

- Optimizer: AdamW (learning rate 1×10^{-4})
- Batch size: 32
- Epochs: 10
- Best model selected based on validation performance

V. GRAD-CAM AND GRAD-CAM++

To understand what the models look at, we generated heatmaps using Grad-CAM. This method highlights image regions that most influence the final prediction. For VGG16, we used the last convolutional layer. For ResNet50, we used the final residual block.

We created heatmaps for:

- VGG16 classification
- VGG16 regression
- ResNet50 classification
- ResNet50 regression

Grad-CAM++ was also implemented as a bonus variant. It uses refined gradient weighting to produce sharper, more focused maps, helping us verify that the models are looking at actual lesions rather than background noise.

VI. RESULTS AND DISCUSSION

A. Classification Results

Table I shows the classification performance. ResNet50 achieved higher accuracy and macro F1-score than VGG16.

TABLE I: Classification Results on Test Set

Model	Accuracy	Macro F1	Best Val F1
VGG16	0.6557	0.4977	0.5200
ResNet50	0.6882	0.5423	0.5141

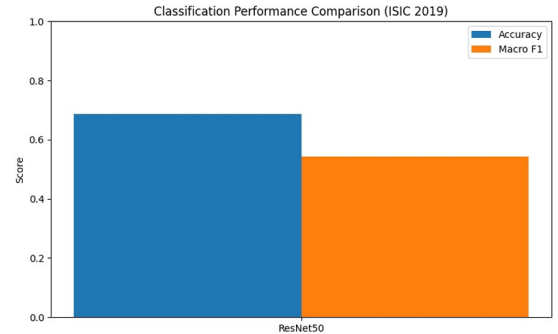


Fig. 2: Visual comparison of accuracy and macro F1-score. ResNet50 consistently outperformed VGG16.

Figure 2 shows the performance difference clearly. The macro F1-score is especially important here because it treats

all eight classes equally, giving a fair view of how well the model handles rare conditions.

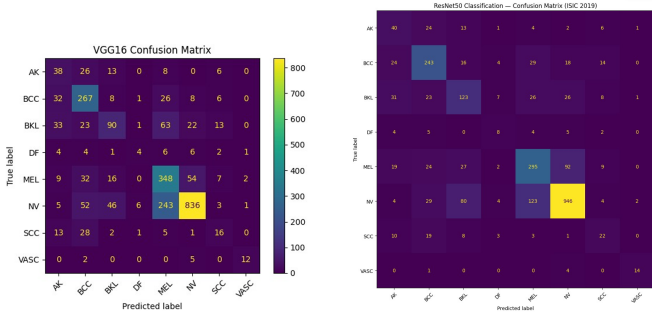


Fig. 3: Confusion matrices for VGG16 (left) and ResNet50 (right). Both models perform well on the common NV class but struggle with rarer types like DF and AK.

Figure 3 shows the confusion matrices. Both models correctly identify most NV cases but have trouble distinguishing rare classes, which is expected given the dataset imbalance.

B. Regression Results

Table II shows the age prediction results. ResNet50 had a lower average error than VGG16.

TABLE II: Age Regression Results on Test Set

Model	MAE (years)	RMSE (years)	R ²
VGG16	10.9142	14.0469	0.3054
ResNet50	10.3367	13.3398	0.3736

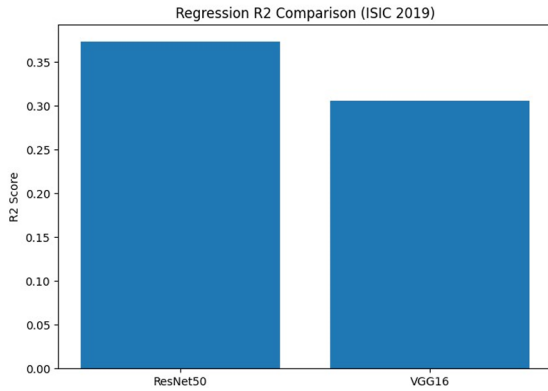


Fig. 4: Comparison of prediction errors. Lower values are better. ResNet50 achieved the lowest MAE and RMSE.

Figure 4 shows the error comparison. An MAE of 10.34 years means the model’s age guesses are, on average, about 10 years off. This is expected because age is not directly visible in skin lesions.

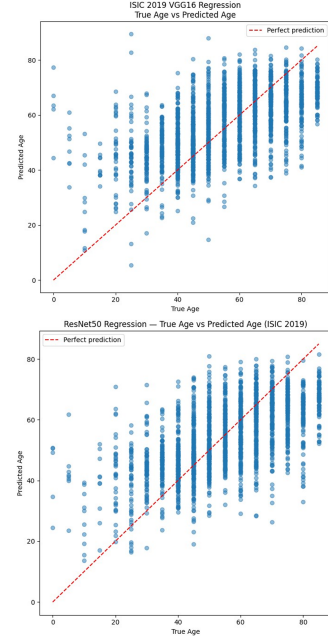


Fig. 5: True vs predicted age scatter plots. Points closer to the red diagonal line are better predictions. Both models capture a general trend but show considerable scatter.

Figure 5 shows the scatter plots. The models follow a general upward trend, confirming they learn some age-related patterns, but the wide spread shows the limits of image-only age prediction.

C. Visual Explanations

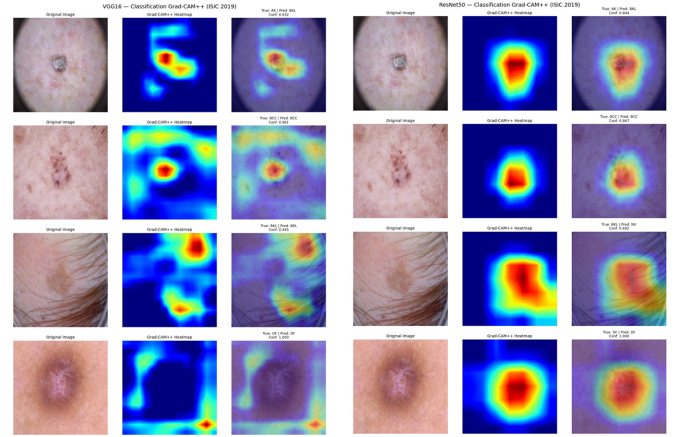


Fig. 6: Grad-CAM heatmaps for classification. Red/yellow areas show model focus. Both models generally highlight the lesion region when predictions are correct.

Figure 6 shows classification heatmaps. When the model predicts correctly, the warm colors concentrate on the lesion. Misclassifications sometimes show attention spreading to background skin, suggesting the model occasionally relies on irrelevant cues.

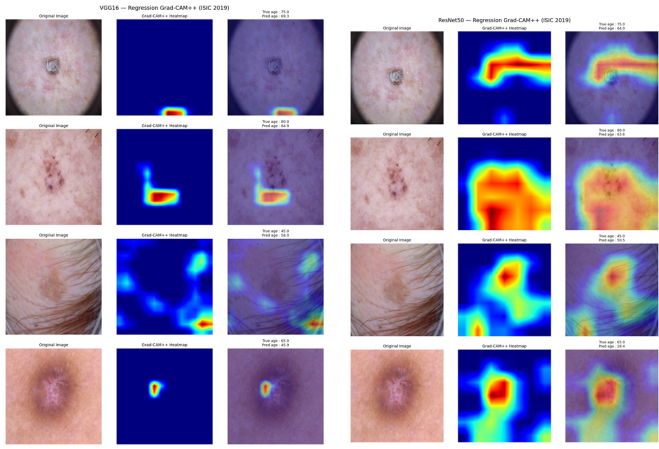


Fig. 7: Grad-CAM heatmaps for age regression. Attention is more scattered compared to classification, supporting the finding that age is harder to learn from images alone.

Figure 7 shows regression heatmaps. The focus is less consistent and more spread out, which matches the quantitative results. Age is influenced by many factors beyond what a lesion image can show.

D. Model Efficiency

ResNet50 used fewer parameters than VGG16 (about 24.6 million vs 40.7 million) while achieving better results. This makes ResNet50 faster to run and more suitable for real-world deployment.

VII. LIMITATIONS AND FUTURE WORK

This study has a few limitations:

- **Class imbalance:** The dataset has many more images of common lesions. Even with class weighting, rare classes remain harder to predict.
- **Age prediction difficulty:** Age is only loosely connected to lesion appearance, so image-only regression will always have limited accuracy.
- **Image-only input:** We did not use other patient data like sex or lesion location. Adding this metadata could improve both tasks.
- **Limited architectures:** Only VGG16 and ResNet50 were tested. Newer models may perform better.

Future work could combine images with clinical metadata, test modern architectures like EfficientNet, use advanced techniques to handle class imbalance, and build a true multi-task network that shares learning between diagnosis and age estimation.

VIII. CONCLUSION

This project built a system that performs both skin lesion diagnosis and age estimation from dermoscopic images. We trained VGG16 and ResNet50 on the ISIC 2019 dataset under identical conditions.

ResNet50 outperformed VGG16 on both tasks:

- Classification: 68.82% accuracy, 0.5423 macro F1-score

- Age regression: 10.34 years MAE, R^2 of 0.3736

Grad-CAM and Grad-CAM++ visualizations confirmed that classification models focus on lesion regions, while regression attention is more diffuse. This matches the idea that diagnosis is easier to learn from images than age prediction.

Overall, ResNet50 proved more accurate and efficient. The results show that deep learning can support multiple medical tasks, though combining images with patient metadata will be key to improving age estimation in the future.

REFERENCES

- [1] "ISIC 2019 Challenge Dataset," International Skin Imaging Collaboration, 2019. [Online]. Available: <https://challenge.isic-archive.com/data/#2019>
- [2] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, p. 180161, 2018.
- [3] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [5] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, pp. 336–359, 2020.
- [6] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2018, pp. 839–847.